

A REVIEW OF TESTS OF VARIOUS HYPOTHESES IN MULTIVARIATE STATISTICAL ANALYSIS

D J de Waal

University of the Orange Free State

Key words: Asymptotic distributions, expansions, canonical correlation, MANOVA, GMANOVA, characteristic roots, covariance matrix, Wishart.

SUMMARY

Various hypotheses in multivariate statistical analysis are considered and references to the asymptotic distributions of the appropriate test statistics in the null and non-null cases given. In most cases the exact distributions can not be obtained in a form which is of practical use and therefore the asymptotic distributions play a very important part in multivariate statistical analysis. Many of these asymptotic expansions are based on the method of box (1949) and can be used for approximate evaluation of significance points, powers and robustness of the various tests. In the cases where the method of Box can not be applied, Taylor series expansions have been done on the characteristic function of the test statistic, see for instance Fujikoshi (1970). Another useful method that is recently used is to express the Wishart matrices into a power series of the sample size (Fujikoshi (1975)). Success has also been achieved by fitting a Pearson distribution (Lec, Krishnaiah and Chang (1976)).

Since tests on the means of multivariate normal populations and a regression analysis can be put in the framework of multivariate analysis of variance (MANOVA), we consider the MANOVA model and some of the important hypotheses on the hypotheses on the parameters. The extension to the general MANOVA model (GMANOVA) that has been given by Potthoff and Roy (1964) is then considered.

Another important testing problem in multivariate analysis is that of independence. This is covered in section 3 on canonical correlations. Tests concerning the covariance matrix are considered in the last section. Tests of this type are important in testing for sphericity, for instance, and in principal component analysis.

1. MANOVA Model

We suppose an $N \times P$ matrix X consists of N independent observations on P variables. The model is (Anderson (1951), Rao (1965), Anderson (1958))

$$(1.1) \quad E(X) = A \Xi$$

where $A : N \times k$ of rank $k \leq N$ is known and Ξ is unknown of order $k \times p$. It is assumed

that X is distributed $N(E(X), \Sigma \otimes I_N)$. The covariance matrix is interpreted as a covariance matrix of the vector formed by stacking the columns of X underneath each other.

We shall consider three hypotheses on this model as given in (1.2), (1.3) and (1.4) below. The first of these is:

$$(1.2) \quad H_{01} : C \Xi D = 0 \quad H_{11} : C \Xi D \neq 0$$

where $C : (u \times k)$ of rank u and $D : p \times v$ are known. Hypotheses of this type cover a wide range of tests. We shall give two examples of tests of this type. Another good example can be found in Smith, Gnanadesikan and Hughes (1962).

Example 1

Let $Y_j^{(i)}$ be an observation from $N(\mu^{(i)}; \Sigma)$, $j = 1, \dots, N_i$; $i = 1, \dots, q$. The null hypothesis is

$$H_0 : \mu^{(1)} = \dots = \mu^{(q)}.$$

Let

$$X' = (Y_1^{(1)}, Y_2^{(1)}, \dots, Y_{N_1}^{(1)}, Y_1^{(2)}, \dots, Y_{N_2}^{(2)}, \dots, Y_{N_q}^{(q)})$$

and $A = (a_{ij})$ $j = 1, \dots, q$; $i = 1, \dots, N$

$$N = N_1 + \dots + N_q$$

with $a_{ij} = \begin{cases} 1, N_1 + N_2 + \dots + N_{j-1} < i \leq N_1 + \dots + N_q; j = 1, \dots, q \\ 0 \text{ otherwise.} \end{cases}$

Let $\Xi = (\mu^{(1)}, \mu^{(2)}, \dots, \mu^{(q)})'$, then

$$E(X) = A\Xi$$

where

X is distributed as $N(E(X); \Sigma \otimes I_N)$.

The hypothesis H_0 is now

$$C \Xi = 0 \Rightarrow (\mu^{(1)} - \mu^{(2)}, \dots, \mu^{(1)} - \mu^{(q)})' = 0$$

where

$$C = \begin{pmatrix} 1 & -1 & 0 & \dots & 0 \\ 1 & 0 & -1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 1 & 0 & 0 & \dots & -1 \end{pmatrix}$$

The following example shows that a regression analysis can also be considered under the MANOVA model.

Example 2.

Suppose we have two groups of individuals $X_1(N_1 \times p)$ and $X_2(N_2 \times p)$ with a regression on A_1 and A_2 respectively.

$$E(X_1) = A_1 \Xi_1, E(X_2) = A_2 \Xi_2$$

$$\text{i.e. } E(X) = A \Xi$$

$$\text{where } X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \quad (N \times p), \quad A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \\ k_1 & k_2 \end{pmatrix} \quad \begin{matrix} N_1 \\ N_2 \end{matrix}$$

$$\text{and } \Xi = \begin{pmatrix} \Xi_1 \\ \Xi_2 \end{pmatrix} \quad \begin{matrix} k \times p, \\ k \times p. \end{matrix} \quad \begin{matrix} N = N_1 + N_2 \\ k_1 = k_2 = k \end{matrix}$$

If we want to test the hypothesis of equality of the last, say s , regression coefficients between the two groups, the hypothesis will be

$$H_0 : \Xi_{12} = \Xi_{22} \quad (k \times s)$$

$$\text{where } \Xi_1 = (\Xi_{11}, \Xi_{12}), \Xi_2 = (\Xi_{21}, \Xi_{22}).$$

The hypothesis H_0 can be written as

$$(I_k, -I_k) \Xi \begin{pmatrix} 0 \ (p-s)s \\ I_s \end{pmatrix} = 0$$

which is of the form (1.2).

Various hypotheses can be considered which are of the general form (1.2).

The second type of hypothesis is

$$(1.3) \quad H_{02} : r(C \Xi D) = r, H_{12} : r(C \Xi D) > r$$

where $r(\cdot)$ denotes the rank of the matrix. An example of an hypothesis of this kind is considered by Anderson (1951).

Example 3

$$\text{Let } \Xi = \begin{pmatrix} \Xi_1 \\ \Xi_2 \end{pmatrix} \quad \begin{matrix} k_1 \times p \\ k_2 \times p \end{matrix}$$

Suppose that the hypothesis $\Xi_2 = 0$ has been tested and rejected. It may however be possible that some of the columns of Ξ_2 are zero. More generally, all the elements of Ξ_2 may be different from zero, but the rank of Ξ_2 may be less than the maximum possible. That implies that it is possible to take a linear combination of components of X such that the expected value of this linear combination is independent of components of A_2 . If the rank of Ξ_2 is r , there are $(k_2 - r)$ linearly independent vectors γ satisfying $\Xi_2 \gamma = 0$.

To test an hypothesis like this, we are testing an hypothesis of the type

$$r(C \Xi D) = r$$

$$\text{where } C = (0, I_{k_2}).$$

The hypothesis $r(C \Xi D) = 0$ reduces to the hypothesis (1.2).

The last type of hypothesis is the type considered by Rao (1973). See also Gleser and Olkin (1966). It is called test for dimensionality.

$$(1.4) \quad H_{03} : C \Xi D \in R_r, H_{13} : C \Xi D \in R_{r'} \text{ for some } r' > r.$$

$\mu \in R_r$ means that each row of the matrix μ actually lies on an r -dimensional plane in the p -dimensional space.

Example 4

$$\text{Let } \Xi = \begin{pmatrix} \Xi_1 \\ \Xi_2 \end{pmatrix} \quad \begin{matrix} k_1 \times p, \\ k_2 \times p \end{matrix}$$

then each row of Ξ_2 is a point in p -dimensional space. Suppose that we want to test the hypotheses:

$$(1.5) \quad H\xi'_{(i)} = w, \quad i = k_1 + 1, \dots, k_2$$

where H is $(r \times p)$ and w is $(r \times 1)$ and $\xi_{(i)}$ is the i -th row of Ξ_2 . H and w are unknown but they are fixed for all i . The hypothesis (1.5) implies that the points $\xi_{(k_1+1)}, \dots, \xi_{(k_2)}$ actually lie on a $(p - r)$ -dimensional plane. The hypothesis (1.5) can therefore be written as

$$C \Xi D \in R_r$$

where

$$C = (0, I_{k_2}) \text{ and } D = I_r.$$

If $w = 0$, H_{03} is equivalent to H_{02} , which implies that if H_{02} is true H_{03} is also true.

We shall now consider each of these three types of hypotheses individually with respect to their test statistics and the asymptotic distribution of the test statistics. We shall first consider a canonical form of the model.

Canonical Form

(See Gleser and Olkin (1966) and Fujikoshi (1974)).

There exist nonsingular matrices T_A, T_C and orthogonal matrices Γ_A, Γ_C such that

$$A = \Gamma_A \begin{pmatrix} I_k \\ 0 \end{pmatrix} T_A$$

$$(1.6) \quad CT_A^{-1} = T_C(I_u, 0) \Gamma_C$$

Transform from X to

$$(1.7) \quad (U, Y) = \begin{pmatrix} \Gamma_C & 0 \\ 0 & I_{N-k} \end{pmatrix} \Gamma'_A X(D_1, D_2)$$

where $D_1 : p \times p - v$ is such that $D_1' D_2 = 0$ and $r(D_1) = p - v$. Then (U, Y) is distributed $N(\cdot; G \otimes I_N)$ where

$$(1.8) \quad E(U, Y) = E \begin{pmatrix} U_1 & Y_1 \\ U_2 & Y_2 \\ U_3 & Y_3 \end{pmatrix} = \begin{pmatrix} \nu_1 & \eta_1 \\ \nu_2 & \eta_2 \\ 0 & 0 \end{pmatrix} \begin{matrix} u \\ k - u \\ N - k \end{matrix}$$

$p-v \quad v$

$$G = \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix} = (D_1, D_2)' \Sigma (D_1, D_2),$$

$$(\nu; \eta) = \begin{pmatrix} \nu_1 & \eta_1 \\ \nu_2 & \eta_2 \end{pmatrix} = \Gamma_C T_A \Xi (D_1, D_2)$$

and

$$(1.9) \quad \eta_1 = T_C^{-1} C \Xi D, D_2 = D.$$

Therefore, tests concerning $C \Xi D$ are now reduced to the distribution of Y . The three hypotheses (1.2), (1.3) and (1.4) now become

$$(1.10) \quad H_{01}^* : \eta_1 = 0, \quad H_{11}^* : \eta_1 \neq 0$$

$$(1.11) \quad H_{02}^* : \eta_1 \in \Pi_r(0), \quad H_{12}^* : \eta_1 \in \Pi_{r'}(0) \text{ for some } r' > r$$

$$(1.12) \quad H_{03}^* : \eta_1 \in \Pi_r(j), \quad H_{13}^* : \eta_1 \in \Pi_{r'}(j) \text{ for some } r' > r$$

where $j = T_C^{-1} 1$, $1 \equiv (1, 1, \dots, 1) : u \times 1$ and $\Pi_r(a)$ has the following meaning:

Let $\Pi_r(a)$ be the set of $u \times v$ matrices M of the form

$$M = GF + ab'$$

for $a : u \times 1$ fixed such that $|G'G| \neq 0$, $FF' = I_r$, and b any $(v \times 1)$ vector.

Then $r(M) = r$ iff $M \in \Pi_r(0)$

$$M \in R_r \quad \text{iff} \quad M \in \Pi_r(1)$$

If Σ is known we can only consider Y_1 , but we assume throughout that Σ is unknown.

We shall now consider each hypothesis individually with respect to its test statistics and their corresponding null and non-null asymptotic distributions.

1.1 Hypothesis (1.2):

$$H_{0I} : C \Xi D = 0, \quad H_{1I} : C \Xi D \neq 0.$$

This hypothesis is well-endowed with test statistics as follows.

(i) LR Criterion

The likelihood ratio test for this hypothesis (Khatri (1966), Gleser and Olkin (1970)), also known as Wilks' criterion, is

$$\Lambda_{11} = \frac{|S_E|}{|S_H + S_E|}$$

where S_E is called the covariance matrix due to error and S_H the covariance matrix due to hypothesis. These are defined as

$$(1.14) \quad S_E = D'SD$$

$$(1.15) \quad S_H = (C \hat{\Xi} D)' M^{-1} C \hat{\Xi} D$$

where

$$(1.16) \quad S = X'(I_N - A(A'A)^{-1}A')X$$

is an estimator for Σ and

$$(1.17) \quad \hat{\Xi} = (A'A)^{-1}A'X$$

is an estimator for Ξ and where

$$(1.18) \quad M = C(A'A)^{-1}C'$$

It is also known that

S_E is distributed as $W(\tilde{\Sigma}, n)$, $n = N - k - (p - q)$, independent of S_H which is distributed as noncentral Wishart $W(\tilde{\Sigma}, u_1, \Omega_1)$

where

$$\tilde{\Sigma} = D' \Sigma D \quad \text{and} \quad \Omega_1 = (C \Xi D)' M^{-1} C \Xi D$$

The asymptotic null distribution of Λ_{11} has been considered by Rao (1948), Box (1949) and is given in Anderson (1958) with a remainder term of $O(n^{-6})$. From Lee, Krishnaiah and Chang (1976) this approximation can be written down to $O(n^{-13})$.

Fujikoshi (1975) derived this asymptotic null distribution to $O(n^{-2})$ in a different way. The non-null asymptotic distribution of Λ_{11} has been derived by Sugiura and Fujikoshi (1969) and Fujikoshi (1975) in terms of noncentral χ^2 variables to $O(n^{-2})$.

(ii) *Hotelling's T_0^2*

It was recently shown by Mudholkar, Davidson and Subbaiah (1974) that Hotelling's T_0^2 , defined as

$$(1.19) \quad \Lambda_{21} = \text{tr } S_H S_E^{-1},$$

is a union-intersection test criterion, a principle introduced by Roy (1957).

The asymptotic null distribution of Λ_{21} has been derived by Ito (1956) and Fujikoshi (1975) to $O(n^{-2})$ in terms of χ^2 variables. Upper percentage points are available in Pearson and Hartley (1972).

The asymptotic non-null distribution of Λ_{21} has been derived by Siotani (1968) to $O(n^{-2})$. Fujikoshi (1970) and (1975) obtained the same result using different methods. The approximations are in terms of noncentral χ^2 variables.

(iii) *Pillai's Statistic*

Pillai (1955) suggested

$$(1.20) \quad \Lambda_{31} = \text{tr } S_H (S_E + S_H)^{-1}$$

as a test criterion for the hypothesis (1.2).

The asymptotic null distribution of Λ_{31} is derived by Fujikoshi (1970) and (1975) in terms of χ^2 variables to $O(n^{-2})$ while the asymptotic non-null distribution of Λ_{31} is given by the same author to $O(n^{-2})$ in terms of non-central χ^2 variables. Values of the upper 5% and 1% points are tabulated by the same author (1970). The results to which we are referring in Fujikoshi (1970) and (1975) are the same, but the methods of proof differ.

Power of Λ_{11} , Λ_{21} , Λ_{31} : The three tests are in general good for testing the hypothesis (1.2). This fact is illustrated by Fujikoshi (1970). It is, however, true that they behave differently for different values of the parameters and that no one of these three can be regarded as best under all values of the parameters.

(iv) *Roy's Largest Root Criterion*

Roy (1957) showed that the largest characteristic root of $S_H S_E^{-1}$ is a union-intersection test criterion for testing (1.2). Tables are available of the exact upper percentage points (Heck (1960)) and associated confidence bounds are available. For example, simultaneous $100(1 - \alpha)\%$ confidence bounds on the function $b'C \Xi Df$ for all $b(u \times 1)$ and $f(v \times 1)$, are given by Roy (1957)) as

$$(1.21) \quad b'C \Xi Df \pm \{[h_\alpha/(1 - h_\alpha)] b'Mb(f'S_E f)\}^{1/2},$$

where h_α denotes the $(1 - \alpha)$ quantile of the distribution tabulated by Heck (1960), with the three parameters s^* , m^* , n^* equal to

$$s^* = \min(u, v)$$

$$m^* = \frac{1}{2}(|u - v| - 1)$$

$$n^* = \frac{1}{2}(N - k - v - 1).$$

In the special case $s^* = 1$, the expression $h_\alpha/(1 - h_\alpha)$ may be replaced by $\frac{(2m^*+2)F_\alpha}{2n^*+2}$,

where F_α denotes the $(1 - \alpha)$ quantile of the F distribution with $(2m^*+2)$ and $(2n^*+2)$ degrees of freedom.

For the purpose of this paper, however, we restrict our attention to Λ_{11} , Λ_{21} and Λ_{31} .

1.2 *Hypothesis (1.3):*

$$H_{02} : \eta(C \Xi D) = r, H_{12} : \eta(C \Xi D) > r.$$

(i) *LR Test*

Anderson (1951) showed that for $D = I_p$:

$$(1.22) \quad \Lambda_{12} = \prod_{j=r+1}^p (1 + d_j)^{-1/2N},$$

where $d_1 > d_2 > \dots > d_p$ are characteristic roots of $S_H S_E^{-1}$, is an LR statistic and derived

the limiting null distribution, i.e. $\lim_{n \rightarrow \infty} -2\ln \Lambda_{12}$ is distributed as $\chi^2_{(p-r)(u-r)}$. Lawley (1959) gave a correction factor to that distribution by considering the asymptotic mean of Λ_{12} under H_{02} . It was, however, proved by Fujikosi (1974), for $D \neq I_p$, but of order $p \times v$, that

$$(1.23) \quad \Lambda_{12} = \{(1 + d_{r+1}) \dots (1 + d_s)\}^{-\frac{1}{2}N}$$

is an LR statistic and that a function of $-2\ln \Lambda_{12}$ approaches a normal distribution under H_{12} . The limiting null distribution of $-2\ln \Lambda_{12}$ given in (1.23) is proved to be (Fujikoshi 1974)

$\chi^2_{(u-r)(v-r)}$. The quantities $d_1 > \dots > d_s$ are the s possibly nonzero roots of $S_H S_E^{-1}$. The proof of these results follows from a result of Hsu (1941) and an application of a theorem by Anderson (1963, p. 140).

1.3 Hypothesis (1.4):

$H_{03} : C \Xi D \in R_p \quad H_{13} : C \Xi D \in R_{p'} \text{ for some } p' > p.$

(i) LR Test

Fujikoshi (1974) has shown that

$$\Lambda_{13} = \{(1 + \ell_{r+1}) \dots (1 + \ell_t)\}^{-\frac{1}{2}N}$$

is an LR test for H_{03} , where $\ell_1 > \ell_2 > \dots > \ell_t$

are the possibly nonzero roots of $S_F S_E^{-1}$ and where

$$(1.25) \quad S_F = (C \Xi D)'(M^{-1} - M^{-1}I(I'M^{-1}I)^{-1}I'M^{-1})C \Xi D.$$

S_F has a noncentral Wishart distribution:

$$W(\tilde{\Sigma}, u-1, \Omega_2), \Omega_2 = (C \Xi D)(M^{-1} - M^{-1}I(I'M^{-1}I)^{-1}I'M^{-1})C \Xi D$$

and $\tilde{\Sigma} = D'\Sigma D$. It is shown by the same author that a function of Λ_{13} has a limiting normal distribution under H_{13} while $-2\ln \Lambda_{13}$ has a limiting $\chi^2_{(u-r)(v-r)}$ distribution.

2. GMANOVA Model

We shall now consider the general MANOVA model given by Potthoff and Roy (1964) namely

$$(2.1) \quad E(X) = A \Xi B$$

where

$X : N \times p$, $A : N \times k$, of rank k , $B : q \times p$ of rank q are known and $\Xi : k \times q$ unknown.

It is assumed that

X is distributed $N(EX; \Sigma \otimes I_N)$.

An example where this model is appropriate is in the analysis of growth curves on several groups of individuals. An excellent introduction to the model with examples can be found in Grizzle and Allen (1969).

Example 5

Consider k groups of experimental subjects with N_j subjects in the j -th group. Each group is subjected to a different treatment. Subjects in all groups are measured at the same p points in time and it is assumed that the covariance matrix is the same at each of the p points. The growth curve associated with the j -th group is

$$\xi_{j0} + \xi_{j1}t + \xi_{j2}t^2 + \dots + \xi_{j,p-1}t^{p-1}$$

Hence

$$A : (\Sigma N_i \times k) = \begin{pmatrix} 1 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \end{pmatrix} \begin{matrix} N_1 \\ \\ N_2 \\ \\ N_k \end{matrix}$$

$$\Xi = (\xi_{j,\ell-1})$$

$$j = 1, \dots, k; \ell = 1, \dots, p.$$

$$B = \begin{pmatrix} 1 & 1 & \dots & 1 \\ t_1 & t_2 & \dots & t_p \\ t_1^2 & t_2^2 & \dots & t_p^2 \\ \dots & \dots & \dots & \dots \\ t_1^{q-1} & t_2^{q-1} & \dots & t_p^{q-1} \end{pmatrix}$$

where $t_1, t_2, \dots, t_p$ are the p points of observation. X is therefore an observation matrix of $N_1 + N_2 + \dots + N_k = N$ by p . The three types of hypotheses considered here, are the same as for the MANOVA case namely $H_{0i}, i = 1, 2, 3$.

Rao (1965) has shown that the model (2.1) is reducible to the usual MANOVA model with covariance adjustment. The reduction is:

Consider the t

Canonical Form

Let $F_1 = \Sigma_0^{-1} B' (B \Sigma_0 B')^{-1}$ where Σ_0 is any positive definite matrix and $F_2 (p \times p - k)$ such that $B F_2 = 0, r(F_2) = p - q$.

Consider the transformation

$$(2.2) \quad (Y^*, Z) = X(F_1, F_2).$$

Then the conditional model of Y^* given Z is written as

$$(2.3) \quad N(A^* \Xi^*; \Sigma_{11.2}^* \otimes I_N),$$

where

$$A^* = (A, Z)$$

$$\Xi^* = (\Xi', R')$$

$$\Sigma_{ij}^* = F_i' \Sigma F_j$$

$$R = \Sigma_{22}^{*-1} \Sigma_{21}^*$$

and

$$(2.4) \quad \Sigma_{11.2}^* = \Sigma_{11}^* - \Sigma_{12}^* \Sigma_{22}^{*-1} \Sigma_{21}^*$$

Since the transformation from Σ to $(\Sigma_{11.2}^*, R, \Sigma_{22}^*)$ is one-to-one and the marginal distribution of Z depends only on Σ_{22}^* , the LR statistic Λ_{1i} for testing H_{0i} , $i = 1, 2, 3$, are equal to the LR statistics obtained by considering the conditional model (1.1).

Representing $C \in D$ as $C^* \in D^*$ where $C^* = (C, 0)$ and noting that the model (2.1) is a special case of (1.1) we can obtain the LR statistics for the GMANOVA model. Let

$$\begin{aligned}
 M^* &= C^*(A^{*'}A^*)^{-1}C^{*'} \\
 \begin{pmatrix} \hat{\Xi} \\ \hat{R} \end{pmatrix} &= (A^{*'}A^*)^{-1}A^{*'}Y^* \\
 S^* &= Y^{*'}(I_N - A^*(A^{*'}A^*)^{-1}A^{*'})Y^* \\
 S_E &= S^*, S_H = (C^* \hat{\Xi} D^*)' M^{*-1} C^* \hat{\Xi} D^* \\
 S_F &= (C^* \hat{\Xi} D^*)' (M^{*-1} - M^{*-1} 1(1' M^{*-1} 1)^{-1} 1' M^{*-1}) C^* \hat{\Xi} D^* \\
 S_H &= (C^* \hat{\Xi} D^*)' M^{*-1} C^* \hat{\Xi} D^*
 \end{aligned}
 \tag{2.5}$$

then from Khatri (1966) in terms of the original matrices

$$\begin{aligned}
 S_E &= D'(BS^{-1}B')^{-1}D \\
 S_H &= (C \hat{\Xi} D)' Q^{-1} C \hat{\Xi} D \\
 S_F &= (C \hat{\Xi} D)' (Q^{-1} - Q^{-1} 1(1' Q^{-1} 1)^{-1} 1' Q^{-1}) C \hat{\Xi} D
 \end{aligned}
 \tag{2.6}$$

where

$$\begin{aligned}
 \hat{\Xi} &= (A'A)^{-1}A'XS^{-1}B'(BS^{-1}B')^{-1}, \\
 Q &= M^* = C [(A'A)^{-1} + (A'A)^{-1}A'X \{S^{-1} - S^{-1}B'(BS^{-1}B')^{-1}BS^{-1}\} \\
 &\quad X'A(A'A)^{-1}]C'
 \end{aligned}
 \tag{2.7}$$

The LR statistics for testing the hypotheses H_{0i} , $i = 1, 2, 3$ can be shown (Fujikoshi, 1974) to be the same as for the MANOVA case by using (2.6) for the matrices due to error and due to hypotheses. Apart from the LR statistic for testing H_{01} , Fujikoshi

(1974) also considered Hotelling's T_0^2 statistic and Pillai's statistic.

Asymptotic Distributions

To obtain the asymptotic distribution of Λ_{1i} , $i = 1, 2, 3$ (Fujikoshi, 1974), take orthogonal matrices Γ_A and Γ_C as defined in (1.6) and make the transformation

$$(2.8) \quad \begin{pmatrix} V_1 & W_1 \\ V_2 & W_2 \\ V_3 & W_3 \end{pmatrix} \begin{matrix} u \\ k-u \\ N-k \end{matrix} = \begin{pmatrix} \Gamma_C & 0 \\ 0 & I_{N-k} \end{pmatrix} \Gamma'_A(Y^*, Z) \begin{pmatrix} D & 0 \\ 0 & I_{p-q} \end{pmatrix}$$

$v \quad p-q$

Then each row of (V, W) is independently distributed normal

$$(2.9) \quad \left(\begin{pmatrix} \eta & 0 \\ 0 & 0 \end{pmatrix} ; \begin{pmatrix} D'\Sigma^*_{11} D & D'\Sigma^*_{12} \\ \Sigma^*_{21} D & \Sigma^*_{22} \end{pmatrix} \right)$$

where $\eta = \Gamma_C T_A \Xi D$. Hence S_E, S_H and S_F defined in (2.5) can be written as

$$(2.10) \quad \begin{aligned} S_E &= V'_3 (I_{N-k} - W_3(W'_3 W_3)^{-1} W'_3) V_3 \\ S_H &= \{V'_1 - W'_1(W'_3 W_3)^{-1} W'_3 V_3\}' \{I_u + W_1(W'_3 W_3)^{-1} W'_1\}^{-1} \\ &\quad \{V_1 - W_1(W'_3 W_3)^{-1} W'_3 V_3\} \\ S_F &= \{V_1 - W_1(W'_3 W_3)^{-1} W'_3 V_3\}' \{W_4^{-1} - W_4^{-1} j G' W_4^{-1} j\}^{-1} j W^{-1} \} \\ &\quad \{V_1 - W_1(W'_3 W_3)^{-1} W'_3 V_3\} \end{aligned}$$

where $W_4 = I_u + W_1(W'_3 W_3)^{-1} W'_1$, $j = T_C^{-1} 1$

It is now easy to verify the following lemma:

Lemma 1

If we consider the distribution of the roots of $S_H S_E^{-1}$ or $S_F S_E^{-1}$, we may assume

that given W_1, W_3 and W_4

- (i) S_E is distributed $W(I_v, n)$
- (ii) S_H is distributed $W(I_v, u, \Delta_1)$
- (iii) S_F is distributed $W(I_v, u-1, \Delta_2)$
- (iv) S_E is independent of S_H and S_F ,

where $n = N - k - (p - q)$,

$$\Delta_1 = L \{I_u + W_1(W_3'W_3)^{-1}W_1'\}^{-1}L',$$

$$L = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_s}, 0),$$

$w_1 \geq \dots \geq w_s \geq 0$ are the possibly nonzero roots of Ω_1 ,

$$\Delta_2 = \{T_C^{-1}C \in D\}' \{W_4^{-1} - W_4^{-1}j(j'W_4^{-1}j)^{-1}j'W_4^{-1}\} T_C^{-1}C \in D$$

or Δ_2 can be written as

$$\Delta_2 = (C \in D)' \{M^{-1} - M^{-1}l(l'M^{-1}l)^{-1}l'M^{-1}\} C \in D + O(n^{-1})$$

2.1 Test (1)

$$H_{0I} : C \in D = 0; \quad H_{1I} : C \in D \neq 0$$

The three test statistics considered here are

$$(2.11) \quad LR : \quad \Lambda_{11} = \frac{|S_E|}{|S_H + S_E|}$$

where S_E and S_H are defined in (2.6)

$$(2.12) \quad \text{HOTELLING'S } T_0^2 : \quad \Lambda_{21} = \text{tr} S_H S_E^{-1}$$

$$(2.13) \quad \text{PILLAI'S STATISTIC} : \quad \Lambda_{31} = \text{tr} S_H (S_H^{-1} + S_E)^{-1}$$

The asymptotic null distributions are the same as given in section 1.1. They can also be

found from the non-null distributions derived by Fujikoshi (1974a) in terms of noncentral χ^2 variables if $\Omega_1 = O(1)$ to $O(n^{-2})$ and in terms of derivatives of normal variables if $\Omega_1 = O(n)$ to $O(n^{-1})$.

2.2 Test (2)

$$H_{02} : r(C \Xi D) = r; \quad H_{12} : r(C \Xi D) > r.$$

The LR statistic for this test has been derived by Fujikoshi (1974), i.e.

$$(2.14) \quad \Lambda_{12} = \{(1 + d_{r+1}) \dots (1 + d_s)\}^{-\frac{1}{2}N}$$

where $d_1 > d_2 > \dots > d_s > 0$ are the possibly nonzero roots of $S_H S_E^{-1}$.

The author has proved the limiting null distribution of $-2\ln\Lambda_{12}$ to be $\chi^2_{(u-r)(v-r)}$ while the limiting nonnull distribution of a function of Λ_{12} is normal.

2.3 Test (3)

$$H_{03} : C \Xi D \in R_r : H_{13} : C \Xi D \in R_{r'} > r.$$

The LR statistic for this test has been derived by Fujikoshi (1974), i.e.

$$(2.15) \quad \Lambda_{13} = \{(1 + \ell_r) \dots (1 + \ell_t)\}^{-\frac{1}{2}N}$$

where $\ell_1 > \dots > \ell_t > 0$ are the possibly nonzero roots of $S_R S_E^{-1}$ and S_F is defined in (2.6).

The limiting null and non-null distributions of Λ_{13} can be obtained on the same lines on which the corresponding distributions for Λ_{12} in (2.14) were derived.

To summarise sections 1 and 2 the following table has been constructed:

TABLE 1

MANOVA

$E(X) = A \Xi$

Hypothesis	Statistic	Asympt. Null Distr.	Asymptotic Non-null Distr.
$C \Xi D = 0$	$LR: \Lambda_{11} = \frac{ S_E }{ S_H + S_E }$	Rao (1948) Box (1949) Anderson (1958) Lee, Krishnaiah & Chang (1976) Fujikoshi (1975)	Sugiura & Fujikoshi (1969) Fujikoshi (1975)

Hypothesis	Statistic	Asympt. Null Distr.	Asymptotic Non-null Distr.
$r(C \Xi) = r$	Hotelling: $\Lambda_{21} = \text{tr} S_H S_E^{-1}$	Ito (1956) Fujikoshi (1975)	Siotani (1968) Fujikoshi (1970) Fujikoshi (1975)
	Pillai: $\Lambda_{31} = \text{tr} S_H (S_H + S_E)^{-1}$	Fujikoshi (1970) Fujikoshi (1975)	Fujikoshi (1970) Fujikoshi (1975)
	Roy: $\max. \text{ch}(S_H S_E^{-1})$ $\Lambda_{12} = \prod_{j=r+1}^p (1+d_j)^{-1/2N}$	Anderson (1951) Lawley (1959)	
$r(C \Xi D) = r$	$\Lambda_{12} = \prod_{j=r+1}^s (1+d_j)^{-1/2N}$	Fujikoshi (1974) (limiting)	Fujikoshi (1974) (limiting)
$C \Xi D \in R_r$	$\Lambda_{13} = \prod_{j=r+1}^t (1+d_j)^{-1/2N}$	Fujikoshi (1974) (limiting)	Fujikoshi (1974) (limiting)

GMANOVA

$$E(X) = A \Xi B$$

$C \Xi D = 0$	$\Lambda_{11} = \frac{ S_E }{ S_H + S_E }$	Fujikoshi (1974) Fujikoshi (1974a)	Fujikoshi (1974a) $\Omega_1 = 0(1), \Omega_1 = 0(n)$
	$\Lambda_{21} = \text{tr} S_H S_E^{-1}$	Fujikoshi (1974) Fujikoshi (1974a)	Fujikoshi (1974a) $\Omega_1 = 0(1), \Omega_1 = 0(n)$
	$\Lambda_{31} = \text{tr} S_H (S_H + S_E)^{-1}$	Fujikoshi (1974)	Fujikoshi (1974a) $\Omega_1 = 0(1), \Omega_1 = 0(n)$
$r(C \Xi D) = r$	$\Lambda_{12} = \prod_{j=r+1}^s (1+d_j)$	Fujikoshi (1974) (limiting)	Fujikoshi (1974) (limiting)
$C \Xi D \in R_r$	$\Lambda_{12} = \prod_{j=r+1}^t (1+d_j)$	Fujikoshi (1974) (limiting)	Fujikoshi (1974) (limiting)

3. Test for Independence

Suppose $X: p \times 1 = \begin{pmatrix} X^{(1)} \\ X^{(2)} \end{pmatrix}$ $q \times 1$ is distributed as

$p - q \times 1$

$N(\mu; \Sigma)$. We want to know if $X^{(1)}$ is independent of $X^{(2)}$. The hypothesis of independence is

$$H_{01} : \Sigma_{12} = 0, \quad H_{11} : \Sigma_{12} \neq 0$$

where
$$\Sigma = \begin{pmatrix} \Sigma_{11}(q \times q) & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

Based on a sample of N observations it is well known that $N\hat{\Sigma} = A$ (say) where $\hat{\Sigma}$ is the ML estimator of Σ and is distributed as $W(\Sigma, n)$, $n = N - 1$. The LR statistic for testing this hypothesis is (Anderson (1958))

$$(3.1) \quad \Lambda_{11} = \frac{|A_{11.2}|}{|A_{11.2} + G|}$$

where

$$\begin{aligned} A_{11.2} &= A_{11} - A_{12}A_{22}^{-1}A_{21} \\ G &= A_{12}A_{22}^{-1}A_{21} \end{aligned} \quad \text{and where}$$

A is partitioned as
$$A = \begin{pmatrix} A_{11}(q \times q) & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

If H_0 is true then $A_{11.2}$ is distributed as $W(\Sigma_{11.2}, n-p+q)$, G is distributed as $W(\Sigma_{11.2}, p-q)$ and $A_{11.2}$ and G are distributed independently of each other.

$\Sigma_{11.2}$ is defined similarly to $A_{11.2}$.

Hence the asymptotic null distribution of Λ_{11} is the same as that discussed in section 1.1. Other statistics for this problem are Hotelling's T_0^2 , $\Lambda_{21} = \text{tr}GA_{11.2}^{-1}$ and Pillai's statistic, $\Lambda_{31} = \text{tr}G(\Lambda_{11.2} + G)^{-1}$. Their corresponding asymptotic null distributions are referred to in section 1.1. Roy's largest root criterion may of course also be used in this situation.

The asymptotic non-null distributions of Λ_{i1} , $i = 1, 2, 3$ have been derived by various authors. They are different from the non-null distributions treated in section 1.1, because (Troskie, 1969) under H_{11} , G conditionally on A_{22} is distributed as $W(\Sigma_{11.2}, p-q, \Omega)$ where $\Omega = \Sigma_{11.2}^{-1/2} \Sigma_{12} \Sigma_{22} \Sigma_{21} \Sigma_{11.2}^{-1/2}$. Let $P = \Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$, then the roots of P are known as the squared population canonical correlations say, $1 \geq \rho_1^2 \geq \dots \geq \rho_q^2 \geq 0$. Essentially two cases are treated under H_{11} , i.e. $P = 0(1)$ which

means that the noncentrality parameter is fixed with respect to n and $P = O(n)$ which means that P is some function of n , say $P = \frac{1}{n} \theta$.

Under a fixed alternative Sugiura and Fujikoshi (1969) derived the asymptotic non-null distribution of Λ_{11} to $O(n^{-1})$ in terms of derivatives of normal distributions. Sugiura (1969) derived the asymptotic non-null distribution of Λ_{11} in terms of non-central χ^2 variables to $O(n^{-2})$ if $P = O(n)$.

Sugiura and Nagao (1973) obtained the asymptotic non-null distributions for Λ_{21} and Λ_{31} if $P = O(1)$ in terms of derivatives of normal distributions to $O(n^{-1})$ while Fujikoshi (1970) derived the asymptotic distributions in terms of noncentral χ^2 variables to $O(n^{-2})$ if $P = O(n)$.

De Waal et al (1975) have obtained the same results as Fujikoshi, using a different approach.

Another interesting test considered by Bartlett (1947) is that of testing the equality to zero of the last $(q-k)$ population canonical correlation coefficients, i.e.

$$H_{02} = \rho_k^2 > \rho_{k+1}^2 = \dots = \rho_q^2 = 0; H_{12} : \rho_{k+1}^2 > \dots > \rho_k^2 > 0$$

These hypotheses are the same as

$$r(\Sigma_{12}) = k; r(\Sigma_{12}) > k.$$

Fujikoshi (1974) proved that

$$(3.2) \quad \Lambda_{12} = \{(1 + r_{k+1}^2) \dots (1 - r_q^2)\}^{-\frac{1}{2}N}$$

is an LR statistic for testing H_{02} against H_{12} where $r_1^2 > r_2^2 > \dots > r_q^2 > 0$ are the squared sample canonical correlations. They are also the characteristic roots of

$$R = A_{11}^{-1} A_{12} A_{22}^{-1} A_{21}.$$

Fujikoshi (1975) also considered

$$(3.3) \quad \Lambda_{22} = \sum_{j=k+1}^q r_j^2 / (1 - r_j^2)$$

and

$$(3.4) \quad \Lambda_{23} = \sum_{j=k+1}^q r_j^2$$

as test statistics. These may be regarded as Hotelling's T_0^2 and Pillai's statistic, respec-

tively. Fujikoshi (1975) also derived the asymptotic null distributions of Λ_{j2} , $j = 1, 2, 3$ in terms of central χ^2 variables to $O(n^{-1})$. Bartlett (1938) has also derived the limiting distribution for Λ_{12} in the null case while Lawley (1959) made some improvements on that. The asymptotic non-null distributions were also derived by him to $O(n^{-1/2})$ in terms of derivatives of normal distributions. To summarize we have the following table:

TABLE 2
Tests for Independence

Hypothesis	Statistic	Asymptotic Null Distribution	Asymptotic Non-null Distribution
$\Sigma_{12} = 0$	LR: $\Lambda_{11} = \frac{ A_{11.2} }{ A_{11.2} + G }$	Same as in Table 1	Sugiura & Fujikoshi (1969) $P = O(1)$
	Hotelling's: $\Lambda_{21} = \text{tr} G_{11.2}^{-1}$		Sugiura (1969) $P = O(n)$
	Pillai: $\Lambda_{31} = \text{tr} G(A_{11.2} + G)^{-1}$		Sugiura & Nagao (1973) $P = O(n)$
$r(\Sigma_{12}) = k$	$\Lambda_{21} = \prod_{j=k+1}^q (1 - r_j^2)$	Bartlett (1938) Lawley (1959) Fujikoshi (1975)	Fujikoshi (1970) $P = O(1)$
	$\Lambda_{22} = \sum_{j=k+1}^q \frac{r_j^2}{1 - r_j^2}$	Fujikoshi (1975)	De Waal, et al (1975) $P = O(n)$
	$\Lambda_{23} = \sum_{j=k+1}^q r_j^2$	Fujikoshi (1975)	Fujikoshi (1975)

4. Tests on equality of the characteristic roots of a covariance matrix

Let $nS: p \times p$ be distributed $W(\Sigma, n)$ and let $d_1 > \dots > d_p$ and $\delta_1 \geq \dots \geq \delta_p$ be the characteristic roots of S and Σ respectively. The hypotheses considered here (for given a and b) are

$$H_0: \delta_{a+1} = \dots \delta_{a+b} = \delta, 0 \leq a \leq a+b \leq p$$

$$H_1 = \text{not } H_0.$$

It is interesting to note that if $a = 0$, $b = p$ then H_0 is the test for sphericity. Therefore H_0 is quite general and is important in principal component analysis, for instance. If we wish to test whether the last b principal components may be ignored, a equals $(p - b)$. Anderson (1963) has considered this problem although Bartlett (1954) and Lawley (1956) were the first to investigate approximations for the test statistics in the null case. The two test statistics are

(i) if δ is known:

$$(4.1) \quad \Lambda_1 = -\ln \prod_{j=a+1}^{a+b} \left(\frac{d_j}{\delta} \right) + \sum_{j=a+1}^{a+b} \left(\frac{d_j}{\delta} \right) - b$$

(ii) if δ is unknown:

$$(4.2) \quad \Lambda_2 = -\ln \prod_{j=a+1}^{a+b} d_j + b \ln \left\{ \frac{1}{b} \sum_{j=a+1}^{a+b} d_j \right\}$$

Anderson (1963) considered the limiting distributions of $n\Lambda_1$ and $n\Lambda_2$ in the null case. Sugiura (1975) has derived an asymptotic expansion for Λ_1 and Λ_2 under H_1 if $\delta_{a+1}, \dots, \delta_{a+b}$ are all simple.

Fujikoshi (1975) obtained the asymptotic null distributions of Λ_1 and Λ_2 to $O(n^{-1})$ in terms of χ^2 variables.

REFERENCES

- ANDERSON, T.W. (1951). Estimating linear restrictions on regression coefficients for multivariate normal distributions. *Ann. Math. Statist.* **22** 327-351.
- ANDERSON, T.W. (1958). *Introduction to Multivariate Statistical Analysis*. John Wiley, New York.
- ANDERSON, T.W. (1963). Asymptotic theory of principal component analysis. *Ann. Math. Statist.* **34** 122-148.
- BARTLETT, M.S. (1938). Further aspects of the theory of multiple regression. *Proc. Bam. Phil. Soc.* **34** 33-40.
- BARTLETT, M.S. (1954). A note on the multiplying factors for various χ^2 approximations. *J. Roy. Statist. Soc., Ser. B.* **16** 296-298.
- BOX, G.E.P. (1949). A general distribution theory for a class of likelihood criteria. *Biometrika* **36** 317-34.
- DE WAAL, D.J., PORTIER, K.M., WEIDEMAN, L. and MAJUMDAR, H. (1975). Asymptotic distributions for Hotelling's T_0^2 and Pillai's statistics for testing independence - a different approach. *Department of Statistics, University of North Carolina, Mimeo Series No. 1001*.

- FUJIKOSHI, Y. (1970). Asymptotic expansions of the distributions of test Statistics in multivariate analysis. *J. Sci. Hiroshima Univ. Serv. A-I*, 34 73-144.
- FUJIKOSHI, Y. (1974). The likelihood ratio tests for the dimensionality of regression coefficients. *J. Mult. Anal.* 4 327-340.
- FUJIKOSHI, Y. (1974a). Asymptotic expansions of the non-null distributions of three statistics in GMANOVA. *Ann. Inst. Statist. Math.* 26 289-297.
- FUJIKOSHI, Y. (1975). Asymptotic expansions for the distributions of some multivariate tests. *Paper presented at the fourth symposium on Multivariate Analysis held in Dayton, Ohio.*
- GLESER, L.J. and OLKIN, I. (1966). A k-sample regression model with covariance. In *Multivariate Analysis I*, Edited by P.R. Krishnaiah, Academic Press, New York, pp. 59-72.
- GLESER, L.J. and OLKIN, I. (1970). Linear models in multivariate analysis. In *Essays in Probability and statistics*. The University of North Carolina monograph series No. 3 in Probability and Statistics.
- GRIZZLE, J.E. and ALLEN, D.M. (1969). Analysis of dose and growth response curves. *Biometrics* 25 357-381.
- HECK, D.L. (1960). Charts of some upper percentage points of the distribution of the largest characteristic root. *Ann. Math. Statist.* 31 625-642.
- HSU, P.L. (1941). On the limiting distribution of the roots of a determinantal equation. *J. London Math. Soc.* 16 183-194.
- ITO, K. (1956). Asymptotic formulae for the distribution of Hotelling's generalized T_0^2 statistics. *Ann. Math.* 27 1091-1105.
- KHATRI, C.G. (1966). A note on a MANOVA model applied to problems in growth curves. *Ann. Inst. Statist. Math.* 18 75-85.
- LAWLEY, D.N. (1956). Tests of significance for the latent roots of covariance and correlation matrices. *Biometrika* 43 28-136.
- LAWLEY, D.N. (1959). Tests of significance in canonical analysis. *Biometrika* 46 59-66.
- LEE, J.C., KRISHNAIAH, P.R. and CHANG, T.C. (1976). On the distribution of the likelihood ratio test statistic for compound symmetry. *South African Statist. J.* 10 49-62.
- MUDHOLKAR, G.S., DAVISON, M.L. and SUBBAIAH, P. (1974). A note on the union-intersection character of some MANOVA procedures. *J. Mult. Analysis* 4 486-493.
- PEARSON, E.S. and HARTLEY, H.O. (1972). *Biometrika Tables for Statisticians, II*, University Press, Cambridge.
- POTTHOFF, R.E. and ROY, S.N. (1964). A general multivariate analysis of covariance model useful especially in growth curve problems. *Biometrika* 51 313-326.
- RAO, C.R. (1948). Tests of significance in multivariate analysis. *Biometrika* 35 58-79.
- RAO, C.R. (1973). *Linear Statistical Inference and its Applications*, Second Edition.

John Wiley, New York, pp. 556-560.

ROY, S.N. (1957). *Some Aspects of Multivariate Analysis*, John Wiley, New York.

SIOTANI, M. (1968). Some methods for asymptotic distributions in multivariate analysis.

Department of Statistics, University of North Carolina, Mimeo Series No. 595.

SUGIURA, N. (1969). Asymptotic non-null distributions of the likelihood ratio criteria for covariance matrix under local alternatives. *Department of Statistics, University of North Carolina, Mimeo Series No. 609.*

SUGIURA, N. (1975). Asymptotic non-null distributions of the likelihood ratio criteria for the equality of several characteristic roots of a Wishart matrix. To appear in *Statistical Papers in Honor of Junciro Ogawa.*

SUGIURA, N. and FUJIKOSHI, Y. (1969). Asymptotic expansions of the non-null distributions of the likelihood ratio criteria for multivariate linear hypothesis and independence. *Ann. Math. Statist.* **40** 942-952.

SUGIURA, N. and NAGAO, H. (1973). Asymptotic formulas for the distributions of two trace statistics for testing independence under fixed alternative. *Comm. in Statist.* **1** 285-295.

TROSKIE, C.G. (1969). The generalised multiple correlation matrix. *South African Statist. J.* **3** 109-121.

Manuscript received 11 March 1976.